Bounded twist relations and retained witnesses

2026-10-30 PR2381 exact finite native study

Actual source and twist. Over $\mathbf{F}_{13}$, $E : y^2 = x^3 + x + 4$, $T : y^2 = x^3 + 4x + 6$, the $d = 2$ monic twist. $P = (0, 2)$ maps to $U = (8, 11)$, both order7, using the prior checked coordinate map Φ and inverse $I = [4]D$ on seven-torsion. No generic rational-point twist isomorphism or target-log lookup is assumed.

Full fixed scope in each instrument. All16 canonical/matched/pullback factor-base arms, original/reduced input, caps0–8:288 configurations, each with six separate unknown-source target attempts. These repeat six points under policies and budgets; they are not1728distinct DLP challenges.

Native independently checked	Flat	Retained
Configurations	288	288
Source-target attempts	1728	1728
Verified original-source recoveries	60	696
Blocked public-log setups	278	172
Capped response queries	1708	1196
Policy/target/cap comparison records	9720	9720
Higher rank / lower cap, recovery admitted	0	192
Beating strongest source policy	0	0
Altered controls rejected	9	9
Backend tests passed	22	22

Observation. Retaining an already computed smaller witness for the *same polynomial* improves certified admission and source recovery under this solver contract. The192records compare policy/target/budget pairs; they are not192independent targets or wins. None survives the strongest-source comparison. No runtime, intrinsic regularity, stationary-yield or full-cost advantage is established.

Failures retained. First Rust build failed duplicate Clone derive before data. First full producer covered288cases but audit rejected applying a two-root-only helper to a three-root base. Correction restores the already specified no-reduction rule. All288producer case bytes reproduce unchanged in the accepted full rerun. Earlier outcomes and protocol amendments are explicit; this is deterministic same-instance replay, not blinded inference.

Source recovery across certified degree budgets

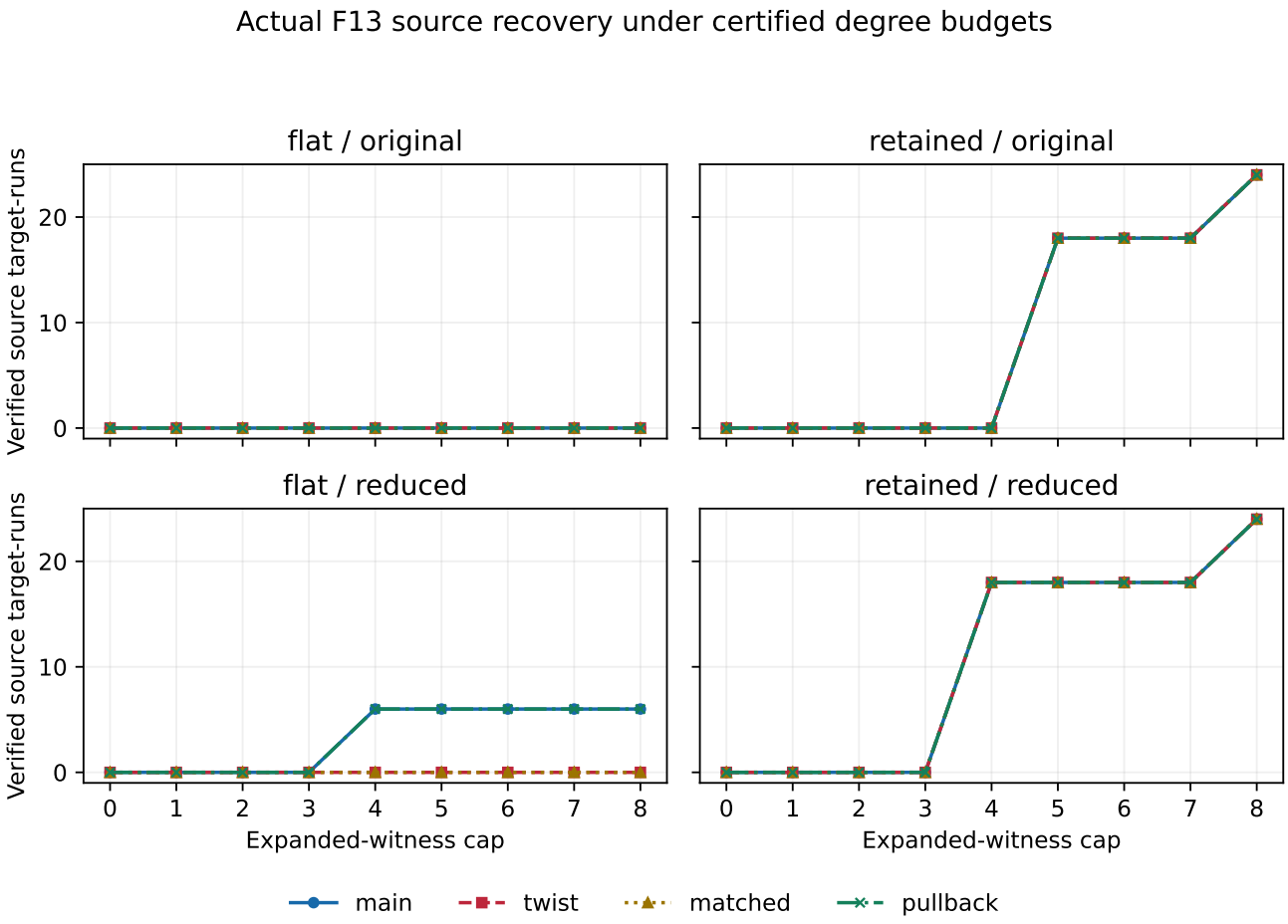
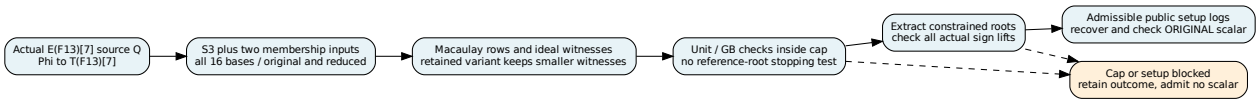


Figure. Native-recorded verified configuration-target runs per cap. Each role contains four policies and six repeated separate targets (maximum24). All values are exact; no sampling or timing uncertainty is inferred. Original and reduced inputs, all losses and matched controls remain separate.

Algebraic track. The solver receives S_3 and both product-membership polynomials. A flat Macaulay row has an explicit witness in these three inputs. Acceptance requires a unit witness or bounded Gröbner certificate before root extraction. A cap failure admits no roots; it does not prove the system empty. Cartesian extraction and actual sign lifts happen only after certification. Direct source enumeration is a separate generic-decomposition reference and has no matrix-degree metric.

Actual bound. Both matrix row degree and expanded original-input witness degree of verification must fit the rung. A shift above the bound is denied before construction. Retained witnesses are local to one solve and are reused only for an identical polynomial, with native ideal expansion checked independently. All matrix work is still executed; a lower certificate bound does not establish lower total cost.

Public setup, original scalar and remaining gates



Only admitted public relations derive base logs; non-full-rank setups remain blocked. Every accepted target relation uses actual unknown Q, all sign lifts and failed shifts, then checks the scalar on original E/P. The prior checked theorem, with explicit public $A_i = \ell_i U$, gives

$$\sum_i v_i A_i = \Phi(Q) + sU, \quad 7Q = 0 \quad \Longrightarrow \quad Q = \left(\sum_i v_i \ell_i - s \bmod 7 \right) P.$$

That theorem and the old18polynomial identities are reused within their domains; no new Lean/global audit or field-identity claim is made.

Evidence. Both full independent native replays pass; each set of nine rehashed controls rejects101without a receipt and the same22backend tests pass in each crate. All6251raw paths restore to exact hashes and repack. Duplicate payloads restore from frozen canonical files; every altered payload remains. Archive37124bytes and compressed manifest35984bytes meet policy. Four canonical exports include both invalid attempts; none is hidden as a successful census.

Limits and next lead. The witness strategy is a finite certification improvement; no strongest-source twist advantage or practical speedup. Setup, field/group work, extraction, verification, allocation and external/rho pricing are not a complete-cost result. Test a new transferable non-isomorphic fixture or encoding against its entire source frontier after source/map verification. No such new fixture search has run here.

Completion hold. Keep PR open, non-draft, auto-merge off. Missing study registrations, naming, incomplete root library suite, required exact-headCI, review, Conductor127/no lease, integration and broader geometry/probability/ regularity/full-work gates remain. No unrelated global gate or frozen evidence changed. Existing local workers only;600second commands inside08:25–09:25UTC; no paid provisioning or arbitrary result upload.

Sources and exact source/hash/flow/export links are in the adjacent README and validation directory. Current protocols precede their affected runs; the failed-audit repair and seen baseline are disclosed. Native witnesses are finite polynomial records, not a universal verified solver theorem.